

2 Subject content

The subject content sets out the knowledge, understanding and skills relevant to this qualification. Together with the assessment information, it provides the framework within which centres create their programmes of study, so ensuring the possibilities for progression to higher education.

Qualification aims and objectives

The aims and objectives of this qualification are to enable students to:

- use creativity and imagination when applying iterative design processes to develop and modify designs, and to design and make prototypes that solve real world problems, considering their own and others' needs, wants, aspirations and values
- identify market needs and opportunities for new products, initiate and develop design solutions, and make and test prototypes
- acquire subject knowledge in design and technology, including how a product can be developed through the stages of prototyping, realisation and commercial manufacture
- take every opportunity to integrate and apply their understanding and knowledge from other subject areas studied during Key Stage 4, with a particular focus on science and mathematics, and those subjects they are studying alongside A Level Design and Technology
- be open to taking design risks, showing innovation and enterprise while considering their role as responsible designers and citizens
- develop intellectual curiosity about the design and manufacture of products and systems, and their impact on daily life and the wider world
- work collaboratively to develop and refine their ideas, responding to feedback from users, peers and expert practitioners
- gain an insight into the creative, engineering and/or manufacturing industries
- develop the capacity to think creatively, innovatively and critically through focused research and exploration of design opportunities arising from the needs, wants and values of clients/end users
- develop an in-depth knowledge and understanding of materials, components and processes associated with the creation of products that can be tested and evaluated in use
- be able to make informed design decisions through an in-depth understanding of the management and development of taking a design through to a prototype
- be able to create and analyse a design concept and use a range of skills and knowledge from other subject areas, including mathematics and science, to inform decisions in design and the application or development of technology
- be able to work safely and skilfully to produce high-quality prototypes
- have a critical understanding of the wider influences on design and technology, including cultural, economic, environmental, historical and social factors
- develop the ability to draw on and apply a range of skills and knowledge from other subject areas, including the use of mathematics and science for analysis and informing decisions in design.

Mathematics

Maths skills are fundamental to design and technology. These will be embedded within the examination for this qualification. Please see *Appendix 1: Mathematical skills requirements* for full details of these.

Scientific skills, knowledge and understanding

Science skills, knowledge and understanding underpin the theory and practice of design and technology. These science skills will be embedded within the examination of this qualification. Please see *Appendix 3: Science knowledge and skills requirements* for full details of these.

Links to other relevant subjects

In addition to maths and science, the A Level GCE in Design and Technology embeds knowledge, techniques and practices from a wide range of subjects including art and design, business, computer science and geography (see *Appendix 4*).

How to use Appendices 1, 3 and 4

As part of the delivery of this qualification, teachers should use these appendices to ensure students gain an awareness and appreciation for the ways in which skills and knowledge from maths, science and other subjects inform decisions in design and the application or development of technology.

Component 1: Principles of Design and Technology

Overview

Students will be required to apply knowledge and understanding of a wide range of materials; including modern and smart materials, and processes used in product design and manufacture. They will be required to develop an understanding of contemporary industrial and commercial practices applied to designing and manufacturing products, and to appreciate the risks involved. Students should have a good working knowledge of health and safety procedures and relevant legislation.

Students must have a sound working knowledge of the use of ICT and systems and control, including modern manufacturing processes and systems, and students will be expected to understand how these might be applied in the design and manufacture of products.

Designers from the past provide inspiration for present and future designing. Students should be aware of the important contribution that key historical movements and figures have on modern design thinking.

It is increasingly important that students develop an awareness of wider issues in design and technology, that design and technological activities can have a profound impact on the environment and on society and that these, together with sustainability, are key features of design and manufacturing practice.

Mathematical and scientific principles are an important part of designing and developing products and students will be expected to apply these principles when considering the designs of others. Please see *Appendix 1: Mathematical skills requirement* and *Appendix 3: Science knowledge and skills requirement*.

Content

Topic	What students need to learn:
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